

XCMG CONSTRUCTION MACHINERY CO., LTD.

SHOP SERVICE MANUAL: XCMG XE215C HYDRAULIC EXCAVATOR

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CHAPTER 1: GENERAL MACHINE SPECIFICATIONS & SAFETY GUIDELINES

1.1 General Description

The XCMG XE215C is a 21.7-metric-ton crawler excavator powered by an Isuzu 4HK1X inline-4 turbocharged intercooled diesel engine producing 133 kW (178 hp) at 2000 RPM. The machine features a negative flow control hydraulic system with tandem Kawasaki K3V112DT variable displacement axial piston pumps and a Nabtesco main control valve.

Operating Weight: 21,700 kg

Standard Bucket Capacity: 0.95 m³

Engine Model: Isuzu 4HK1X (Tier 3 / Stage IIIA compliant)

Hydraulic System Operating Pressure: 34.3 MPa (343 bar)

Travel Speed (High/Low): 5.5 / 3.3 km/h

Swing Speed: 11.8 RPM

Fuel Tank Capacity: 360 Liters

Hydraulic Oil Tank Capacity: 220 Liters (Full System: 320 Liters)

SAFETY WARNING:

Before servicing any hydraulic lines or accumulator circuits, lower all work equipment to the ground, stop the engine, turn the starter key to ON, and cycle control levers in all directions to release residual hydraulic pilot pressure.

SECTION 2: HYDRAULIC SYSTEM TROUBLESHOOTING & PRESSURE SETTINGS

2.1 Main Pump & Pressure Relief Diagnostics

The hydraulic system uses twin variable displacement axial piston pumps (Pump 1 & Pump 2) providing a maximum rated flow of 2 x 216 L/min.

SYMPTOM: BOOM CYLINDER DRIFT OR WEAK DIGGING FORCE

When the boom cylinder drifts downward under static load exceeding 60 mm in 15 minutes, perform the following inspection sequence:

1. Connect a 0-600 bar calibrated digital pressure gauge to Pilot Check Port M1.
2. Warm the hydraulic oil to 50C - 60C (ISO VG 46 high VI anti-wear oil).
3. Measure Main Relief Valve (MRV) cracking pressure at engine high idle (2000 RPM).
4. If pressure is below 31.0 MPa (310 bar), adjust pilot relief spring shim or replace valve cartridge P/N 803182104.
5. If MRV pressure is normal (34.3 MPa) but boom drifts, inspect boom holding valve spool for contamination or worn check seal P/N 803201452.

SPECIFICATION TABLE 2.1: HYDRAULIC PRESSURE & TORQUE RATINGS

Component / Test Point	Standard Spec	Tolerance Limit	Part Number / Notes
Main Relief Valve (Working)	34.3 MPa (343 bar)	33.5 - 35.0 MPa	Valve Cartridge: 803182104
Power Boost Setting	36.3 MPa (363 bar)	35.5 - 37.0 MPa	Active 8 sec duration
Pilot System Pressure	3.9 MPa (39 bar)	3.7 - 4.1 MPa	Pilot Valve: 803164821
Boom Holding Valve Pressure	36.3 MPa (363 bar)	35.0 - 37.0 MPa	Seal Kit: 803201452
Swing Motor Relief Pressure	26.5 MPa (265 bar)	25.5 - 27.5 MPa	Brake Valve: 803175210
Drive Sprocket Mounting Bolts	270 Nm (199 lbf-ft)	+/- 20 Nm	M18 Class 10.9 (Torque Spec)
Slewing Ring Bearing Bolts	560 Nm (413 lbf-ft)	+/- 35 Nm	M20 Class 10.9 (Torque Spec)
Track Roller Mounting Bolts	190 Nm (140 lbf-ft)	+/- 15 Nm	M14 Class 10.9 (Torque Spec)

SECTION 3: ENGINE DIAGNOSTICS & SAE J1939 FAULT CODES

3.1 Isuzu 4HK1 Electronic Engine Diagnostics

The Isuzu ECM communicates with the machine display via SAE J1939 CAN bus network at 250 kbps baud rate.

SPN 110 FMI 0: ENGINE COOLANT TEMPERATURE CRITICALLY HIGH (FLASH CODE 14)

- Threshold: Engine coolant temperature exceeds 105C (221F) for > 10 seconds.
- Probable Root Causes:
 1. Radiator cooling pack clogged with dust/debris.
 2. Thermostat stuck in closed position (P/N 8980170270).
 3. Water pump impeller slippage or drive belt tensioner failure.
 4. Viscous fan clutch failure or low coolant level.
- Corrective Repair Steps:
 1. Clean radiator cooling fins using low-pressure compressed air (< 0.2 MPa).
 2. Inspect drive belt tension; verify tensioner arm alignment mark.
 3. Replace thermostat assembly (cracking temp: 82C, full open: 95C).
 4. Tighten water pump housing bolts to 26 Nm.

SPN 100 FMI 1: ENGINE OIL PRESSURE CRITICALLY LOW

- Threshold: Engine oil pressure drops below 0.8 bar (12 PSI) at low idle or 2.1 bar at high idle.
- Probable Root Causes:
 1. Low engine oil level or incorrect viscosity grade (use CI-4 / CK-4 15W-40).
 2. Oil filter bypass valve jammed open (Filter P/N 8980188580).
 3. Main crankshaft bearing excessive clearance.
- Corrective Repair Steps:
 1. Shut down engine immediately to prevent catastrophic crankshaft seizure.
 2. Inspect oil level on dipstick; check for fuel dilution (smell or elevated level).
 3. Replace lube oil filter element 8980188580, torque to 38 Nm.

SPN 651 FMI 5: INJECTOR CYLINDER #1 CURRENT BELOW NORMAL / OPEN CIRCUIT

- Threshold: Open circuit detected on Common Rail Injector Cylinder #1 wiring harness.
- Corrective Repair Steps:
 1. Measure injector solenoid resistance with digital multimeter: specification is 0.4 to 0.8 ohms at 20C.
 2. Inspect engine wiring harness connector ECM pin E-12 for moisture or pin corrosion.
 3. If injector resistance is > 1.2 ohms, replace common rail injector P/N 8980116045.